Data bibliography / attribution

Orbital Technosphere / Observer

01 CelesTrak / orbital elements and memberships

26 OMM CSV catalogs. Collected 2026-11-03T17:13:12Z to 2026-11-03T17:14:05Z.
Active catalog supplies the satellite population; membership catalogs identify named fleets. Three bounded debris catalogs supplement it.
Catalogs: active, starlink, oneweb, kuiper, qianfan, hulianwang, gnss, iridium-next, globalstar, orbcomm, resource, science, stations, fengyun-1c-debris, iridium-33-debris, cosmos-2251-debris, planet, spire, intelsat, ses, eutelsat, telesat, gps-ops, glo-ops, galileo, beidou.
Attribution: CelesTrak. Exact feed URLs, element epochs and hashes are retained in the source ledger.
https://celestrak.org/NORAD/documentation/gp-data-formats.php

02 SGP4 / Vallado, Crawford, Hujsak and Kelso

Positions propagated from the pinned elements using the reference SGP4 implementation, WGS-72 and AFSPC mode.
Calculation instant: 2026-11-03T17:10:00Z. 44 records excluded by epoch; 0 propagation failures.
Public element uncertainty and approximate Earth rotation limit these visualization positions.
https://celestrak.org/publications/AIAA/2006-6753/

Historical validation / NASA SSCWeb

Reference instant: 2026-08-13T15:36:00Z.
Historical regression reference only. The November refresh was unreachable; this response does not provide an independent comparison of the plotted November positions.
https://sscweb.gsfc.nasa.gov/WebServices/REST/

Cartography / Cartofreako

Projection: Authagraph. The projection bibliography credits its mathematical and design sources.
Separate map and legend composition follows the Setouchi ST2 layout. Typography: Atkinson Hyperlegible; bundled fonts under the SIL Open Font License.
https://bdekoz.github.io/cartofreako/docs/pages/projections/authagraph/bibliography.html

Pinned inputs / source ledger

The profile records the calculation time, observer, filters and every catalog URL. Adjacent acquisition.json and SHA256SUMS record retrievals and original input bytes.
Map and legend carry the same timestamp and counts; the complete acquisition ledger is embedded in this SVG.
Map image scaled to 98.863636% with proportional sheet margins.
https://github.com/bdekoz/cartofreako/blob/main/assets.static/orbital-technosphere/20261103/orbital-technosphere-profile.json

Setouchi AuthaGraph: +4.35-inch printed horizontal move on a 43.5-inch image. Celestial directions retain source registration in the same cyclic frame.

MSP speculation 4 / palette 2 / Ultramarine / porcelain

Randoma11y: ground #193fb8; type #f5eae7. sRGB contrast 7.28:1 (screen only). Vendor markers retain source colors.
https://randoma11y.com/%23193fb8/%23f5eae7
Epson Stylus Pro 9900 / metallic glossy paper / cyan and light magenta reported low; grays reportedly print violet.
Intended RGB hues; final printed hue is unmeasured. No green correction, ink separation or metallic reflection simulation.

SGP4 positions / 2026-11-03T17:10:00Z

1,088 plotted objects / named fleets and residual roles

Seven-day element age; six-hour future tolerance.

- Starlink 549
- OneWeb 44
- Kuiper 31
- Qianfan 20
- Hulianwang 21
- Iridium 5
- Globalstar 4
- ORBCOMM 0
- Planet 0
- Spire 2
- Intelsat 38
- SES 26
- Eutelsat 7
- Telesat 12
- GPS 11
- GLONASS 8
- Galileo 13
- BeiDou 13
- Debris 53
- Other active 220
- Other navigation 2
- Other Earth observation 6
- Science 2
- Human presence 1

One object per NORAD ID. Overlapping named memberships use profile order.

Name and locpoint sublayers are independent. Names are hidden by default.

Black x: debris. Sunburst: Starlink. Square: Intelsat. Triangle: remaining Earth observation.

Active-marker opacity records illumination. Debris remains solid black.

Topocentric celestial directions above the San Francisco horizon. Equator and horizon are reference lines.

Starlink sunbursts: 1.5× linear size on both orbital maps.

Orbital technosphere Observer / Authagraph